

Homework 9

Grant Talbert

November 7, 2025

Boston University

Problem 5

Problem 6

Problem 6.1. Let R be a commutative ring with subset $S \subseteq R$ closed under multiplication and containing 1. Define the relation $\sim$ on $R \times S$ by, for all $s, s' \in S$ and $a, a' \in R$:

$$(a, s) \sim (a', s') \iff \exists t \in S \text{ s.t. } t(s'a - sa') = 0.$$

Prove that $\sim$ is an equivalence relation.

Proof. For this problem, let $a, a', a'' \in R$ and $s, s', s'', t, u \in S$. Given $(a, s) \sim (a', s')$, say the relation is witnessed by t if

$$t(as' - a's) = 0.$$

Of course the witness is not necessarily unique; it is simply convenient terminology.

(reflexivity) By commutativity of R ,

$$1(as - sa) = as - as = 0.$$

Therefore, $(a, s) \sim (a, s)$.

(symmetry) Let $(a, s) \sim (a', s')$ be witnessed by t . Then by commutativity of R ,

$$t(as' - a's) = 0 \iff (-t)(a's - as') = 0 \iff (a', s') \sim (a, s),$$

where the last implication follows by definition of $\sim$. Therefore, $\sim$ is symmetric.

(transitivity) Let $(a, s) \sim (a', s')$ and $(a', s') \sim (a'', s'')$ be witnessed by t and u , respectively. Then

$$t(as' - a's) = 0 = u(a's'' - a''s').$$

Since $u, s'' \in S$, we have $us'' \in S$, so we can write

$$(us''t)(as' - a's) = 0.$$

a

■

Problem 6.2. Denote by $\frac{a}{s}$ the equivalence class $[(a, s)]_{\sim}$. Let $S^{-1}R = (R \times S)/\sim$ be endowed with the operations

$$\frac{a}{s} + \frac{a'}{s'} = \frac{as' + a's}{ss'}, \quad \frac{a}{s} \cdot \frac{a'}{s'} = \frac{aa'}{ss'}$$

for all $a, a' \in R$ and $s, s' \in S$. Show that $S^{-1}R$ is a commutative ring.

Proof. It is immediate by definition that $S^{-1}R$ is closed under both operations. Since R is commutative,

$$\frac{a}{s} + \frac{a'}{s'} = \frac{as' + a's}{ss'} = \frac{a's + as'}{s's} = \frac{a'}{s'} + \frac{a}{s}.$$

We can also show associativity of addition:

$$\begin{aligned} \left(\frac{a}{s} + \frac{a'}{s'}\right) + \frac{a''}{s''} &= \frac{as' + a's}{ss'} + \frac{a''}{s''} = \frac{(as' + a's)s'' + a''ss'}{ss's''} = \frac{ass'' + a'ss'' + a''ss'}{ss's''}, \\ \frac{a}{s} + \left(\frac{a'}{s'} + \frac{a''}{s''}\right) &= \frac{a}{s} + \frac{a's'' + a''s'}{s's''} = \frac{as's'' + (a's'' + a''s')s}{ss's''} = \frac{as's'' + a's's'' + a''s's}{ss's''}. \end{aligned}$$

Invoking commutativity of R shows these quantities are equal. Since S contains $1_R \in S$, define 1 in $S^{-1}R$ as $1_R/1_R$. Then

$$\frac{1_R}{1_R} \cdot \frac{a}{s} = \frac{1_R a}{1_R s} = \frac{a}{s}$$

since 1_R is the identity in R . To show commutativity, we note that since R is commutative,

$$\frac{a}{s} \cdot \frac{a'}{s'} = \frac{aa'}{ss'} = \frac{a'a}{s's} = \frac{a'}{s'} \cdot \frac{a}{s}.$$

The final thing we must do is deal with zero. Define 0 as any element of the form $0/s$; we must show these all fall within the same equivalence class. Indeed, if $(0, s), (0, s')$ are two elements of $R \times S$, then

$$(0s' - 0s) = 0 - 0 = 0,$$

so we have $(0, s) \sim (0, s')$. Since the definition is now well-defined, we observe that

$$\frac{a}{s} + \frac{0}{s'} = \frac{as' + 0s}{ss'} = \frac{as'}{ss'} = \frac{a}{s} \cdot \frac{s'}{s'}.$$

It suffices to show that $(s', s') \sim (1_R, 1_R)$. Once again,

$$(s'1_R - 1_R s') = s' - s' = 0,$$

so we indeed have $s'/s' = 1$. Therefore,

$$\frac{a}{s} \cdot \frac{s'}{s'} = \frac{a}{s} \cdot 1 = \frac{a}{s}.$$

0 is thus an additive identity. Given a/s , define the inverse $-a/s$ by $[(-a, s)]_{\sim}$. Then

$$\frac{a}{s} + \frac{-a}{s} = \frac{-as + as}{s^2} = \frac{0}{s^2} = 0.$$

Therefore, $S^{-1}R$ is a commutative ring. ■

Problem 6.3. Show $S^{-1}R \cong 0$ if and only if $0 \in S$.

Proof. It suffices to show that $(a, s) \sim (0, s')$ for all $(a, s) \in R \times S$ and some $s' \in S$. Consider $s' = 0$. We have

$$(a0 - 0s) = 0 - 0 = 0,$$

so indeed $(a, s) \sim (0, 0) = 0$. Therefore, $S^{-1}R$ is the zero ring. ■

Problem 7